Wearable Electrotactile Feedback for Motor Skill Acquisition

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Abstract

This research investigates embodied learning through electrotactile feedback as a means to teach motor skills in engineering and computing education. Many students in classroom settings struggle to learn dextrous skills such as soldering because direct instructor guidance is limited. Haptic feedback offers a promising avenue to supplement visual and verbal instruction through direct stimulation of neural pathways. The proposed work focuses on developing unobtrusive, wearable electrotactile systems that can scale to classroom use while remaining customizable for instructors. A technical contribution lies in advancing closed-loop electrotactile feedback by exploring electromyography (EMG) based muscle synergy analysis. In a pedagogical sense, the research aims to create scalable, embodied learning interventions that improve motor skill acquisition.

CCS Concepts

• **Human-centered computing → Interactive systems and tools**; • **Hardware** → *Emerging interfaces*; • **Social and professional topics → Computational science and engineering education**; • **Applied computing → Computer-assisted instruction**.

Keywords

Electrotactile feedback, Haptics, Embodied learning, Motor skill instruction, Muscle synergy analysis

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1 Context and Motivation

Haptic feedback is an important aspect of human-computer interaction (HCI) particularly in applications where engaging the body's nervous system is needed to convey information, enhance realism, or replicate environmental cues. Broadly, haptics in HCI can be categorized into (1) cutaneous feedback, which stimulates the skin through vibration, temperature, or pressure, and (2) kinesthetic feedback, which targets pathways associated with force, position, and joint movement.[4, 17]. Conventional haptic devices such as vibrotactile motors or mechanical exoskeletons have been effective but remain constrained by scalability, form factor, and power requirements. Electrotactile (ET) feedback offers a promising alternative. By directly stimulating sensory nerves through patterned electrical signals, it can produce localized tactile sensations without using bulky actuators [17]. Compared to mechanical actuation, electrotactile interfaces are lightweight, unobtrusive, and energy-efficient, with the potential for high spatial resolution and customizable design [14]. On the other hand, a similar methodology, Electrical Muscle Stimulation (EMS), focuses on inducing force feedback via muscle contractions, but a common theme from the user surveys on works in EMS is that the involuntary muscle contractions are rather uncomfortable and disorienting. This makes electrotactile stimulation a good candidate to explore wearable, unobtrusive, and relatively comfortable haptic feedback.

My motivation for exploring electrotactile haptics stems from teaching experience. Many students in tertiary education (e.g., university) struggle to acquire dexterous motor skills like soldering in introductory electronics courses. Large class sizes limit individualized instruction, leaving students reliant on verbal explanations or demonstrations. In one of the courses I taught to computer engineering students, I observed that only a fraction of students reached proficiency under these constraints. Prior work shows that kinesthetic and tactile experiences are central to motor learning [10], yet most educational technologies stop short at audiovisual feedback or basic vibration. This gap motivates my investigation into whether electrotactile feedback can enrich skill acquisition by providing unobtrusive, customizable guidance directly through the nervous system.

The significance of this research is twofold. On the technical front, it advances haptic feedback paradigms by exploring electrotactile stimulation as a lightweight, precise, and scalable modality. On the educational front, it evaluates whether such feedback can support embodied learning at scale, helping students acquire fine-motor skills more effectively in settings where individualized instruction is impractical. Together, these contributions aim to bridge the divide between emerging haptic technologies and their integration into real-world teaching and learning contexts.

2 Background

Electrotactile (ET) feedback delivers electrical current through skin-surface electrodes to stimulate *afferent nerve endings*[1], producing tactile sensations. Unlike electrical muscle stimulation (EMS), which activates motor neurons to induce movement, ET primarily excites cutaneous *mechanoreceptors*. These include Meissner's corpuscles (low-frequency vibration), Merkel's cells (pressure and texture), Pacinian corpuscles (high-frequency vibration), and Ruffini endings (skin stretch) [8]. By selectively or jointly activating these receptors, ET can generate a wide variety of tactile experiences.

[1] Afferent nerves carry sensory information from the environment to the brain.

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In HCI, ET emerged as an alternative to vibrotactile and electromechanical feedback, offering unique benefits: no moving parts, high spatial resolution, and the ability to elicit localized, configurable sensations. Early work from 1991 demonstrated its use for sensory substitution, like tactile vision substitution systems [7]. These studies demonstrated that with training, users could interpret abstract ET signals as functional information, laying the foundation for modern ET interfaces. More recent systems have applied ET for immersive interaction and guidance. For example,

(1) **Notifications:** Compact actuation to provide private cues without moving muscles.
(2) **Wearables:** Duente et al. [3] demonstrated ET feedback on the wrist via smartwatch integration. Similarly a commercial product Afference Phantom [5] is expected to recreate realistic touch sensations through on-skin ET stimulation, signaling industry-level interest in this modality.
(3) **Haptic Rendering in AR/VR:** Works like TelePulse [6] explored biomechanical simulation–based EMS feedback in VR, showing how electrical stimulation can enhance immersion. Similar approaches may extend ET rendering for fine tactile cues in mixed reality.
(4) **Skill acquisition:** Studies have applied ET stimulation to training tasks [12, 18], broader comparative studies also benchmark it against EMS and vibrotactile feedback [16].

Among these applications, a particularly promising area is *motor skill acquisition*, where haptic feedback can augment visual and verbal instruction. Chellali et al.'s WYFIWIF[2] paradigm [2] showed that haptic channels accelerate gesture learning by conveying information that is otherwise difficult to express linguistically.

This connection aligns with the broader paradigm of *embodied learning*, which argues that cognition is shaped through the body's sensorimotor engagement with the world and that meaning-making is grounded in movement and perception rather than abstract reasoning alone [1]. In this view, tactile and kinesthetic interaction is not supplementary to learning but constitutive of it, a position that challenges traditional "mentalistic" models of instruction, which have historically downplayed the role of the body in knowledge construction [9].

In computing education, this perspective is particularly relevant. Many core engineering and computer science (CS) tasks from typing and debugging with specialized hardware to soldering and circuit assembly require fine motor skills that are typically taught through apprenticeship-style instruction. ET feedback offers a scalable, wearable modality to reinforce these skills, providing learners with tactile cues even outside instructor contact.

Despite this promise, several challenges remain:

(1) Limited studies of stimulation sites beyond the fingertips.
(2) Tedious calibration, with no adaptive closed-loop systems yet; EMS-based approaches (e.g., muscle synergy analysis for golf putting [11]) suggest possible directions.
(3) Poor wearability: most ET studies still rely on bulky medical-grade stimulators. Recent miniaturization of EMS hardware into smartwatch form factors [15] hints at similar opportunities for ET systems.

[2]What You Feel Is What I Feel

3 Problem Statement

> Can unobtrusive, closed-loop electrotactile haptic feedback systems improve learning of fine motor skills (e.g., soldering) in computing education given large classroom settings?

This can be considered challenging because,

- Motor learning varies widely across individuals, requiring customizable feedback.
- Effective learning requires keeping the instructor in the loop, not replace them.
- Technically, most existing systems are open-loop and lack adaptability to user responses.

4 Research Goals

My goals are to:

(1) Develop wearable electrotactile prototypes that can be used in a classroom setting. They should be customizable and have tunable stimulation parameters.
(2) Create tools and interfaces that allow instructors to easily configure and program the system without deep technical expertise.
(3) Design closed-loop feedback mechanisms using EMG-based muscle synergy analysis to have adaptive feedback.
(4) Evaluate the pedagogical impact of electrotactile feedback in classroom-based motor skill instruction.

5 Research Methods

- **Prototype development:** Initial proof-of-concept systems adapted from EMS toolkits (e.g., Let Your Body Move EMS toolkit by University of Hannover [13]) to electrotactile modalities.
- **Exploratory experiements:** Open-loop electrotactile feedback to identify optimal electrode placements and stimulation strategies.
- **Closed-loop integration:** Use EMG-based muscle synergy analysis to dynamically adapt feedback.
- **User studies:** Controlled classroom studies with computing/engineering students performing simple soldering tasks. Performance will be evaluated across the first few phases of learning, including retention after rest periods.

6 Contributions

Expected contributions include:

- A new electrotactile haptic system optimized for unobtrusive, classroom-based use.
- A methodology for closed-loop electrotactile feedback leveraging muscle synergy analysis.
- Empirical evidence on the role of electrotactile haptics in scalable embodied learning for motor skill instruction.

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